TRANSACTION ANALYSIS IN DEREGULATED POWER SYSTEMS USING GAME THEORY

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Abstract -- The electric power industry is in transition to a deregulated marketplace for power transactions. In this environment, all power transactions are made based on price rather than cost. A regional power pool (we refer to it as a Pool throughout this paper) is noted as the most straightforward path to a deregulated electricity marketplace. However, many questions remain unanswered regarding the economics of Pool participation. In a deregulated energy marketplace, participants are interested in maximizing their own profits, regardless of the system-wide profits. It is perceived that the competition will reduce the price of electricity for retail customers, however, the key issue for participants is related with the price definition to remain competitive. In this paper, we use the game theory to simulate the decision making process for defining offered prices in a deregulated environment. The outcome of this study may be used by Pool coordinators to discourage unfair coalitions. A modified IEEE 30 bus system is used as a deregulated power pool to illustrate the main features of the proposed method.

Keywords-- Power System Operation, Deregulated Power Pools, Spot Pricing, Game Theory, Network Constraints.

I. INTRODUCTION

As deregulation in the electric industry is becoming a reality, some questions related to the ability of deregulated power pools to conduce the system to maximum efficiency operation require urgent answers. In a fully competitive environment, more efficient power producers and distributors will maximize their revenues, while those offering higher prices may have to improve their capabilities in order to lower the cost of power production and/or delivery. A regional power pool (we refer to it as a Pool in this paper) may be used as a viable path to competition, as physical laws that govern electricity require a human coordinator to support the competitive market. In this environment, tasks related with the provision of transfer capacity, as well as control, coordination and dispatch of power are performed by an independent Pool coordinator (e.g., ISO in the California System).

We assume all Pool participants use a price curve, rather than a cost curve, to exchange power. Participants consider market prices which maximize their benefits, while Pool coordinators try to maximize the system-wide benefits [1-4]. The role of Pool coordinators is clear in this case: transactions that would lead to maximum revenues should be encouraged while those situations in which participants take advantage of bottlenecks in the Pool are to be identified and corrected.

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We use a methodology based on the game theory [5-8] to simulate and analyze the economic behavior of Pool participants. The Pool is modeled as a strategic game in which participants *play* against each other in order to maximize their own benefits. The *strategy* that participants follow is the bid for pricing transactions. From the point of view of Pool coordinators, the methodology presented in this paper is summarized as:

- Identify players of the game and their possible strategies
- Identify possible coalitions among participants
- Compute transactions and economic benefits associated with any coalition
- Identify those coalitions which are likely to be formed
- Encourage those coalitions that would maximize Pool benefits

Network constraints are considered in the formulation. A modified IEEE 30 bus system [17] is used to illustrate the proposed framework for market pricing simulation.

II. TRANSACTIONS IN DEREGULATED ENVIRONMENTS

The generation cost is represented as $C(\mathrm{i}) = a(\mathrm{i}) + b(\mathrm{i})P(\mathrm{i}) + c(\mathrm{i})P(\mathrm{i})^2$, with a linear incremental cost function. Correspondingly, price curves for each generator in participating utilities are defined by a straight line with slope $m(\mathrm{i})$ as in Figure 1. The linear price curve introduces non-linearities in the problem, however, it is a more realistic representation of price than that of a fixed price for power.

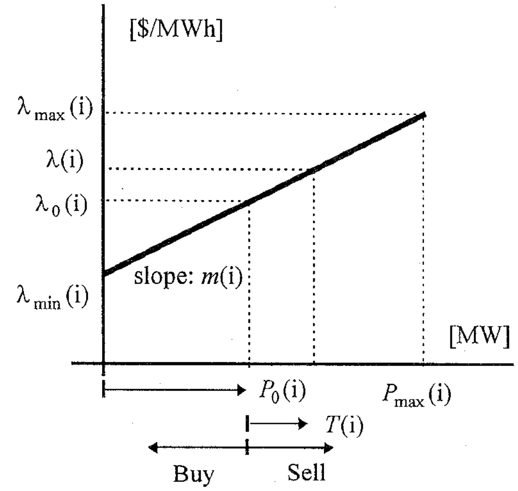

Figure 1: *Price Offer in generator i*

In Figure 1, participants increase the generation beyond $P_0(\mathrm{i})$ if the market price is greater than $\lambda_0(\mathrm{i})$, i.e. the marginal cost of electricity at the generation level $P_0(\mathrm{i})$. $T(\mathrm{i})$ is the net interchanged power; if $T(\mathrm{i})$ is positive (negative), the participant is selling power to (buying power from) the Pool, $-P_0(\mathrm{i}) \leq T(\mathrm{i}) \leq P_{max}(\mathrm{i}) - P_0(\mathrm{i})$. Each generator will produce $P(\mathrm{i}) = P_0(\mathrm{i}) + T(\mathrm{i})$ to supply loads. In this model, Pool participants interact by means of *price* signals, while the cost coefficients are shared only on a voluntary basis. Once the transactions are set, transmission charges are computed as a percentage of revenues.

II.1 Power transactions as a strategic game

We use the game theory [9-13] which attempts to analyze the players' behavior and provide advice for improving the results in a competitive market.

Each participant is introduced as a *player*, economic benefits constitute *payoffs* and players' options are treated as *strategies*. Any subset of players is called a *coalition*, with a *grand coalition* representing all participants. Two classes of games are considered in this paper: a *non-cooperative* game in which any coalition is maximizing its own rewards regardless of the results for the counter-coalition. Inside the coalition, strategies are coordinated among participants in a *cooperative* game. In other words, in non-cooperative games, players make no commitments to coordinate their strategies; conversely, in cooperative games players can make *credible* commitments to coordinate their strategies. In a non-cooperative game, some players choose a strategy as other players try to identify their best response to that strategy. In a cooperative game, players define strategies that lead to the best outcome for a coalition.

There are two sets of agreements discussed by players in a coalition. The first one is to agree on the coordination of strategies, *e.g. we all bid high.* The second one is to agree on the payoffs distribution. In this paper, we do not analyze the second agreement: we assume if a strategy is coordinated for a coalition, the players will find a way to share the payoffs in order to make the coalition possible.

II.2 Coalitions among participants

In Figure 2, the decision process for pricing is given in a deregulated energy marketplace where participation in a Pool is being debated. Players may consider either to *participate* in the Pool or to define *bilateral agreements* with other players. Players may in essence join the Pool if the corresponding economic benefit is higher than that of bilateral agreements.

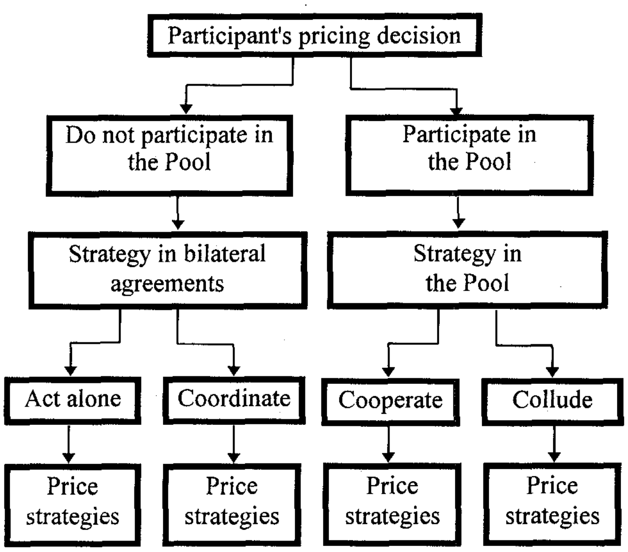

Figure 2: *Participant's pricing decision*

In the case of bilateral agreements, a participant may consider to *act alone* (hence, not coordinating pricing policies with other participants) or to *coordinate* its price strategy with other participants to increase benefits. In our model, the coordination of price strategies is permissible when the pool participation is disregarded, and independent utilities consider bilateral agreements.

In the case of Pool participation, participant may consider either to *cooperate* with the Pool (hence, defining prices that maximize system-wide benefits) or to *collude* with other participants over system-wide benefits. Pool regulations should prevent any coalitions other than the grand coalition, and encourage *cooperation* among Pool members.

The situation when all participants cooperate with the Pool (i.e. grand coalition) conduces to maximum system-wide benefits. In this paper, we simulate the decision making process and identify situations in which the grand coalition is most appealing to players.

The decision making process is analyzed from a "pessimistic" point of view. Each participant defines its strategy assuming the worst economical situation in the market. For instance, when a player considers to act alone, the worst situation is that other players collude to minimize the player's benefits.

III. SIMULATING THE MARKET USING GAME THEORY

III.1 Transactions based on the spot price

When network losses are not considered, the spot price of electricity [14,15] is defined as:

$$\rho = \frac{\partial C}{\partial P_i} + \sum_{\ell=1}^{N_L} \mu_\ell \frac{\partial Z_\ell}{\partial P_i} \qquad \text{for bus i in the Pool}$$

where,

ρ Spot price of electricity
C Total generation cost
P_i Generation level in bus i
N_L Number of lines in the system
μ_ℓ Lagrange multiplier for maximum capacity of line ℓ
Z_ℓ Power flow in line ℓ

In a deregulated system, generation cost is treated as confidential; however, the spot price of electricity may be computed by searching for the minimum price offered in the market that satisfies load and generation constraints [14]. When network constraints are not active, multipliers μ_ℓ are null and ρ coincides with the equal price criterion. After transactions are defined, each seller receives $\rho T(i)$ for generating excess power and each buyer pays $-\rho T(i)$ for imports. The economic benefits of participant "r" is expressed as:

$$\begin{aligned} Benefit(\mathrm{r}) = \sum_{i \in \Omega} \{ & [a(\mathrm{i}) + b(\mathrm{i}) P_0(\mathrm{i}) + c(\mathrm{i}) P_0(\mathrm{i})^2] - \\ & - [a(\mathrm{i}) + b(\mathrm{i}) P(\mathrm{i}) + c(\mathrm{i}) P(\mathrm{i})^2] + T(\mathrm{i})\rho \} \end{aligned} \tag{1}$$

where Ω is the set of generators for participant r. Other pricing schemes [16] would result in different allocation of benefits among participants; however, our approach is general enough to be used in other scenarios as well. These issues are being investigated and will be presented in our future publications.

III.2 Example system

The power system used in this paper is a modified IEEE 30 bus system [17]. We assume that each player is a utility that can either supply its local load and possibly sell power to the pool depending upon the market price. The system topology is shown in Figure 3, where tie lines are drawn with thicker lines.

Before transactions are defined, each utility supplies its local load by applying an economic dispatch. Then, each utility offers prices to the pool using curves similar to that in Figure 1. The characteristics of generators and generation levels without transactions are listed in Table 1. In Table 1, the price for no transactions $\lambda_0(i)$ is the marginal cost evaluated in $P_0(i)$. Based on prices in Table 1, utilities A and C sell power in the grand coalition while utility B buys power.

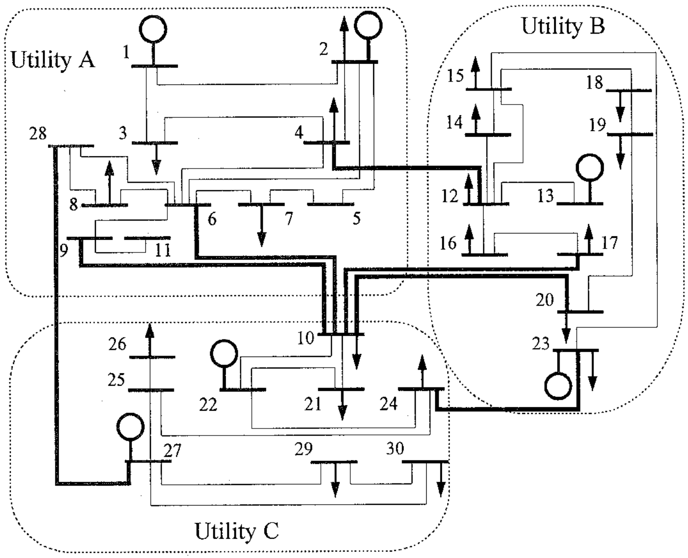

Figure 3: *Example system*

Table 1: *Generator data*

	bus	Cost coefficients $a(i)$ $b(i)$ $c(i)$	Min Max [MW]	$P_0(i)$ [MW]	$\lambda_0(i)$ [\$/MWh]
A	1	0 2 0.02	0 80	23.54	2.94
	2	0 1.75 0.0175	0 80	60.97	3.88
B	13	0 3 0.025	0 40	37.00	4.85
	23	0 3 0.025	0 30	19.20	3.96
C	22	0 1 0.0625	0 50	21.59	3.699
	27	0 3.25 0.00834	0 55	26.91	3.699

IV. NUMERICAL RESULTS AND DISCUSSIONS

IV.1 Strategies of players

Participants are able to change their prices by adjusting the slope in Figure 1. Using constrained economic dispatch, Pool benefits will be maximized when all participants trade power at marginal cost, $m(i)=2c(i)$. As participants try to maximize their *own* benefits, they may either decrease their bids in order to sell more power or increase the price in order to earn more. Among the infinite set of feasible alternatives for each participant, we analyze the following three strategies:

H- Trade power at 1.15 times the marginal cost, $m(i) = 2.3c(i)$. The participant's strategy is to *bid high.*

M- Trade power at marginal cost, $m(i) = 2c(i)$. The participant's strategy is to *cooperate* with the Pool.

L- Trade power at 0.85 times the marginal cost, $m(i) = 1.7c(i)$. The participant's strategy is to *bid low* .

In each case, benefits are positive representing the sum of payments and incremental costs as in (1). In the example system, there are three possible coalitions among participants plus the grand coalition. Correspondingly, there are four possible non-cooperative games between coalitions and counter-coalitions.

IV.2 Transactions in a perfect competition

Let's assume first that no capacity limits are imposed on tie lines and participants are not facing any additional constraints (e.g., interrupible power from contracts, contingencies in local resources, etc.) in defining offered prices. We refer to this as a *perfect competition*, as participants cannot take advantage of external constraints. In the example, no participant has a biased control over the market price because of its size or location .

In Table 2, the payoff matrix of the non-cooperative game is shown for coalition $\mathcal{S}=\{A,B\}$ and its counter-coalition $\mathcal{S}^C=\{C\}$. In this game, utilities A and B agree to join forces in order to obtain higher benefits by combining strategies against utility C. The set of strategies available in each coalition is the product of the strategies for each participant in the coalition. Participants in the coalition have nine possible strategies: both bid at marginal costs (MM), both bid high (HH), both bid low (LL) in addition to (HM, HL, MH, ML, LH, and LM). An entry in the matrix is a pair of payoffs for coalition members and the counter-coalition, respectively. Each entry in the payoff matrix is computed using the spot price to define transactions among utilities for that particular strategy (set of prices). The first value in the pair represents the sum of benefits in utilities A and B. The second value is the benefit obtained by utility C for the same combination of strategies.

Table 2: *Payoff matrix [\$/h]*

{A-B} ↓ {C} ↦	H	M	L
HH	20.30,0.27	20.34, 0.24	20.38, 0.20
HM	20.38,0.42	20.41, 0.38	20.45, 0.32
HL	19.97, 0.65	19.98, 0.59	19.99, 0.56
MH	20.56, 0.18	20.60, 0.16	20.63, 0.14
MM	20.62, 0.30	20.65, 0.28	20.69, 0.23
ML	20.17, 0.50	20.19, 0.46	20.21, 0.39
LH	20.46, 0.10	20.49, 0.09	20.53, 0.07
LM	20.48, 0.19	20.52, 0.17	20.56, 0.14
LL	19.95, 0.34	19.98, 0.32	20.01, 0.27

While individual elements of the matrix are available to corresponding utilities, the entire matrix is accessible to the Pool coordinator.

The *characteristic function* is an estimate of the best possible outcome in the worst situation for the coalition. In this example, the characteristic function of the coalition {A-B} is found by first locating the minimum benefit for coalitions in each row of Table 2 (i.e., \$/h 20.3, 20.38, ... , 19.97) and then choosing the maximum of selected minimums: v(A-B) = \$/h 20.62. The chosen strategy is the *max-min strategy*. In this regard, utilities A and B bid at marginal costs because the bid offers the highest benefit when the other Pool participant (i.e. utility C) is minimizing the coalition's benefits. A similar analysis is made for utility C with v(C) = \$/h 0.1; in this case, the max-min strategy for utility C is to bid high. A linear programming procedure for characteristic function is discussed in the Appendix.

The characteristic function is a pessimistic estimate because it assumes that the counter coalition is playing to minimize the coalition's benefit, when in fact the counter coalition is trying to maximize its own benefits. For instance, in Table 2 if utility C decides to play M instead of its max-min strategy, the benefits of the coalition are higher than the characteristic function.

Other criteria rather than the max-min criterion could be used to simulate the decision process in participants. For instance, both the pessimism-optimism index criterion and the criterion based on the principle of insufficient reason [8] would concentrate the decision in a weighted combination of the best and the worst states for the participants. The choice of the adequate decision criteria will depend upon the characteristics of the actual participants playing the game in the Pool.

The strategy MM for coalition {A-B} in Table 2 is said to be a *dominant* strategy because the coalition chooses the same strategy for different circumstances (strategies of the counter coalition) that the player faces. From Table 2 we learn that the row MM corresponds to the maximum benefit that coalition {A-B} obtains for each column; hence, no matter which strategy utility C chooses, the best strategy for coalition {A-B} is MM. Accordingly, the strategy H is a dominant strategy in utility C.

If all bids are based on characteristic functions, the Pool will not reach the maximum system-wide benefit (i.e. $/h 20.93). For this analysis, utilities A and B will bid at marginal costs while utility C will bid high. The game is in equilibrium at this point, because the strategies are the best response to the opponent's strategy. The combination of dominant strategies (MM and H) and the corresponding payoffs ($/h 20.62 and 0.30) are the *dominant strategic equilibrium* for the game. However, it is a rather non-stable equilibrium because there are other possible coalitions in which utilities may obtain higher benefits (see the Appendix for mathematical derivations of the game equilibrium).

Table 3 gives the characteristic functions of coalitions in a perfect competition. The values for {A-B} and {C} are extracted from Table 2. All other entries are computed from the respective payoff matrices in the 4 possible games.

Table 3: *Characteristic functions [$/h], conditions of perfect competition*

	$v(S)$	$v(S^C)$
{A-C}	9.23	11.35
{A-B}	20.62	0.1
{B-C}	11.81	8.75
{A-B-C}	20.93	-

The grand coalition (fourth row in Table 3) is optimum from the Pool's perspective: load is supplied at minimum cost using available resources, and the maximum system-wide benefits are obtained. This solution is the same as that of a traditional centralized dispatch with minimizing cost functions. In this coalition, participants are obtaining $/h 9.11, 11.54 and 0.28, respectively.

An coalition of utilities that is dominated through some other coalition would never become permanently established. There would be a tendency for the existing coalition to break up and be replaced by one that gives its members a larger share.

Based on Table 3, we can infer which coalitions are likely to form. Clearly, none of the participants alone are doing as good as the grand coalition. For instance, utility B will obtain at least $/h 11.35 by playing against the possible coalition of utilities A and C (first row in Table 3), however, in the grand coalition utility B will obtain $/h 11.54. Therefore utility B, acting rationally, will probably decide to cooperate with the Pool. It is observed also that there are no economic reasons for any of the participants to desert the grand coalition and join other coalitions for higher benefits. This is also the case in Table 2 in which the game between {A-B} and {C} is in a dominant strategy equilibrium; however, coalition {A-B} is a nonstable coalition because its members are able to obtain higher benefits when joining the grand coalition. Mathematically, the allocation in the grand coalition is an imputation (satisfies A.1 and A.2 in Appendix) which represents the core of the game.

It is often claimed that the increasing pressure from competition in the energy market will help maximize the customers' benefits in a pool. This is supported by the results obtained from the game theory. In general, the bigger the game, the greater the variety of coalitions. The grand coalition of the game will be dominating as long as the market is not giving any participants or group of participants a relative advantage over the remaining participants. Hence, in the case of a game played by numerous participants there will be a narrower margin for possible coalitions against system-wide benefits.

IV.3 Transactions in an imperfect competition

Several factors can alter the perfect competition described in the previous section, including the economic pre-eminence of some of the participants, the mix of generation resources available to each participant, the geographic situation of the participants, etc. When the market is in *imperfect competition*, some participants may find that it is possible to obtain higher benefits by colluding with a group of participants.

In this section, we analyze the role of network constraints in a deregulated power pool. We suppose the line between buses 27 and 28 (linking utilities A and C in Figure 3) is on outage. In the previous case, utility C was exporting power using this line. It is cheaper now for utility C to import power than to supply the load locally. In this case, utility A is selling more power and utility B is buying less power in comparison with the case when the line was available. The characteristic functions for this case corresponding to all feasible coalitions are shown in Table 4.

Table 4: *Characteristic functions [$/h], line 27-28 not available*

	$v(S)$	$v(S^C)$
{A-C}	2.79	15.66
{A-B}	18.38	0.19
{B-C}	16.61	1.88
{A-B-C}	18.70	-

As line 27-28 is out, total generation costs are higher and the maximum system-wide benefit is $/h 18.70. In the grand coalition, participants obtain $/h 2.71, 15.83 and 0.16 respectively. In Table 4, utility C is doing better alone (bid high in this case) than joining the grand coalition. From the second row in Table 4 utility C is guaranteed to obtain at least $/h 0.19 when playing against the possible coalition of utilities A and B. This amount is higher than the benefit obtained by utility C in the grand coalition (i.e. 0.16); hence, there is an incentive for utility C to defect the grand coalition.

The coalition of utilities B and C is also likely to form. The strategy for coalition {B-C} in the third row of Table 4 is to allow B to bid below the marginal cost and C to bid above the marginal cost. Utilities B and C may find out that as they coordinate their bids, they obtain at least \$/h 16.61 which is higher than that of the grand coalition. Hence, utilities B and C may decide to coordinate bids and share the extra benefits. In this case, utility A's benefit and that of the system-wide are lower than those obtained in the grand coalition.

If we compare Tables 3 and 4, we learn that when line 27-28 is available, utilities B and C are obtaining the best result in the grand coalition (i.e. \$/h 11.82). Hence, it will be advantageous for utilities B and C if line 27-28 is unavailable. In this case, even when *system-wide* benefits are lower, the individual benefits are higher.

The Pool coordinator may use the procedures described in this section to identify unanticipated problems and take adequate corrective measures.

V. CONCLUSIONS

Deregulation in the electric power industry is expected to increase the benefits associated with the operation of interconnected power systems. In this paper, we use game theory concepts to simulate the behavior of participants in deregulated energy markets. We assume a totally deregulated power pool in which each participant defines prices to interchange power with the remaining participants of the electricity marketplace.

The spot price of electricity is used to define transactions among participants. A modified version of the IEEE 30 bus system is used as an example for deregulated power pools.

From the results obtained in the paper, we foresee that in a perfect competition, all participants try to maximize their benefits by cooperating with the power pool to obtain the maximum system wide benefits. The notion that increasing competition will help decrease operation costs appears to be supported by the results obtained in the paper. Mathematically, increasing competence (i.e. more participants, additional tie lines available to avoid flow congestion, etc.) makes the grand coalition more appealing to all participants.

In this paper, we present a case when conditions for a perfect conditions are altered. The network imposes additional constraints on the bids, and participants increase their benefits by coordinating bid strategies and sharing benefits. Some participants may even prefer a more constrained network as they can take advantage of the situation and obtain higher benefits.

The analysis may be used by Pool coordinators to identify non-competitive situations and to encourage pricing policies that lead to maximum system-wide benefits. Participants in deregulated power pools can also use the specific aspects of the proposed analyses for price definition and decision make processes.

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BIOGRAPHIES

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APPENDIX: GAME THEORY CONCEPTS

We present the theoretical aspects of the game theory which are followed by participants in a deregulated energy market.

A.1 Equilibrium in non-cooperative games

A N-player game *normal form* [4] consists of a set of N players, N strategy sets $\boldsymbol{X}_i$, i=1,...,N and the N-tuple payoff function $\pi(\boldsymbol{X}_1,..., \boldsymbol{X}_N)$. The value $\pi_i(x_1, \ldots , x_N)$ is the payoff function in participant i when player 1 plays the mixed strategy $x_1 \in \boldsymbol{X}_1, \ldots$, player N plays the strategy $x_N \in \boldsymbol{X}_N$. We only consider *pure* strategies. Each player can only choose one strategy.

An N-tuple of strategies $x_1, \ldots , x_N$ is an *equilibrium N-tuple* if

$$\pi_i(x_1, \ldots ,y_i , \ldots , x_N) \leq \pi_i (x_1,..., x_N)$$

for all i and for all strategies y_i for player i; hence, the player departing from the mixed strategy in the N-tuple at least does no better.

In a two-player game, the two-dimensional set $\Pi=\{\pi_1(X_1,X_2), \pi_2(X_1,X_2)\}$ is called the *non-cooperative payoff region* [4]. The points in Π are called *payoff pairs*. If (u,v) and (u',v') are two payoff pairs, (u,v) *dominates* (u',v') if $u \geq v$ and $v \geq v'$.

Payoff pairs which are not dominated by any other pair are said to be *Pareto Optimal*. Clearly rational cooperating participants in a power pool would never play so that their payoffs are not Pareto optimal.

A.2 Characteristic function

The *max-min value* or *characteristic function* gives each player a pessimistic estimate of how much payoff can be expected. In a two-player game, the max-min value gives player 1 (the row player) the expected payoff by assuming that player 2 (the column player) will act to minimize player 1's payoff.

$$v_1 = \max_{X_1} \min_{X_2} \pi_1(X_1, X_2)$$

where X_1 and X_2 range over all mixed strategies for players 1 and 2 respectively. The max-min value for player 2 is defined in a similar way.

The characteristic functions of the coalition and counter-coalition are computed as follows. Let m_{ij} be an element of the row's player payoff matrix. The row's player problem is to compute the row value:

$$v_r = \min_j \sum_{i=1}^{m} p_i m_{ij}$$

and an optimal mixed strategy [4] $\boldsymbol{p} = (p_1, \ldots , p_m)$ with Σp_i=1.

A constant c is added to all entries large enough that $m_{ij}+c>0$ for all ij. The problem is formulated now as a function of variable $y_i = p_i / v$ for $1\leq i \leq m$; the problem is formulated then as a linear programming:

$$\text{minimize } y_1+\ldots+y_m$$

$$\text{subject to } \sum_{i=1}^{m} m_{ij} y_i \geq 1 \quad \text{for } 1 \leq j \leq n$$

The column's player problem in the non-cooperative game is computed in the same way after transposing the column's player payoff matrix.

In a two participant game, the assumption that participant 2 will try to minimize 1's benefit is probably false. Participant 2 will try instead to maximize his benefits; however, v_1 gives participant 1 a lower bound (pessimistic estimate) on his benefit.

A.3 N-players cooperative games

As indicated above, a *coalition* is a subset of players which is formed in order to coordinate strategies and to agree on how the total payoff is to be divided among its members.

The set of all N players is denoted by $\mathcal{P}$. Given a coalition $\mathcal{S} \subseteq \mathcal{P}$, the *counter-coalition* to $\mathcal{S}$ is:

$$\mathcal{S}^c = \mathcal{P} - \mathcal{S} = \{\mathcal{P}_i \in \mathcal{P} : \mathcal{P}_i \notin \mathcal{S}\}$$

In general, in a game with N players, there are 2^N coalitions.

The game between any coalition $\mathcal{S}$ and its counter-coalition $\mathcal{S}^c$ is a non-cooperative game. The rows in the matrix correspond to the set of strategies available to $\mathcal{S}$ and the columns correspond to those available to $\mathcal{S}^c$. The max-min value for the coalition is called the *characteristic function* of $\mathcal{S}$ and is denoted $v(\mathcal{S})$.

When a coalition is formed, payoff is distributed among players in the coalition. The amounts going to each player form a N-tuple X of real numbers. Two conditions are usually required from an N-tuple in order to be likely to actually happen in the game [4]:

- *individual rationality*

$$x_i \geq v(\{\mathcal{P}_i\}) \tag{A.1}$$

- *collective rationality*

$$\sum_{i=1}^{N} x_i = v(\mathcal{P}) \tag{A.2}$$

Individual rationality requires that the coalition gives any player a higher payoff than that the player can obtain on his own. A N-tuple of payments which satisfied both these conditions is called an *imputation.*

The *core* of a game consists in all imputations which are not dominated by any other imputation through any coalition. Hence, if an imputation X is in the core, there is no group of players which has a reason to form a coalition and replace X with a different imputation.

Every imputation in the core of the game is an *efficient* allocation. The standard definition of efficient allocation is Pareto optimality, if the allocation is efficient, there is no player who can do better without making some other player worse off.

Mathematically X is in the core if and only if:

$$\sum_{\mathcal{P}_i \in \mathcal{S}} x_i \geq v(\mathcal{S})$$

for every coalition $\mathcal{S}$.

The concept of core of the game has an important limitation: the core of the game may be any size; it may even be empty, meaning that there are no stable coalitions. Whatever coalition is formed, there is some incentive for a subgroup to desert the coalition.

In a deregulated power system as presented in this paper, the objective from the Pool coordinator's perspective should be to make the distribution of savings among participants to be in the core of the game for the case of the grand coalition.

Discussion

J. Jacobs (Pacific Gas & Electric Co., San Francisco, CA) Ferrero et al. have produced an interesting study of the way participants in a deregulated power market will likely behave. although I find the stated conclusions to be less interesting than some other inferences.

In examining the paper's examples one must bear in mind that "supply" and "demand" are placed on an equal footing. This is different from our normal models of monopoly and monopsony. The "grand coalition" includes all parties. buyers as well as sellers. The goal of a grand coalition is to maximize the total benefit, often called "social welfare". irrespective of the allocation of those benefits. With convex production functions. welfare is maximized by pricing at marginal cost.

The authors use the term "perfect competition" to describe a set of constraints and production functions for which the grand coalition is in the core of the associated multiplayer game. In that case all players will set prices at marginal cost. This is not the same as the normal use of the term "perfect competition" for a situation where all participants are price takers. In fact one of the contributions of this paper is the detailed analysis of an example of imperfect competition (in the usual usage).

In the absence of the assurance of collusion, utility C in the example of IV.2 would likely assume the worst (that A and B are in league against it). In that case the authors show that C would be more likely to price significantly (15%) above cost than at marginal cost. Collusion may be a necessary condition for marginal cost pricing.

The example of IV.3 then shows that even if players were able to collude, it may not be in their interest (self-interest may be as powerful a weapon against the grand coalition as antitrust enforcement). One can demonstrate that the conditions of individual and collective rationality for IV.3 have no feasible solution. The stable coalitions must involve pricing at other than marginal cost. The authors lead the reader to think that the grand coalition fails only because a normally operative intertie is down, but the cause is irrelevant.

Many participants in debates over electricity restructuring assume that market players will price at marginal cost. This paper gives two counters to that assumption. First, marginal cost pricing is a property of the grand coalition, which may only be establishable by explicit collusion. Absent such collusion optimizing players might choose other strategies. Second, network constraints might make the grand coalition unstable. It may be necessary to introduce entirely new supplies to assure marginal cost pricing. Of course, the ultimate goal of industry restructuring is not just improved pricing today but the rational introduction of new supplies (long-term efficiency, not short-term).

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Hugh Rudnick, David Watts (Universidad Católica de Chile, Santiago, Chile) and **Diego Moitre** (Universidad Nacional de Rio Cuarto, Rio Cuarto, Córdoba, Argentina). The authors are to be congratulated for their paper, which uses a simple game theory approach to characterize and simulate a competition environment in the electric generation sector and where perfect and imperfect competition conditions are assessed. Strategies that could be chosen by individual firms to price their energy offers are analyzed (below, equal to or over their marginal costs). The paper only deals with short term competition at the spot wholesale market and does not analyze competition that takes place on the long term contract market.

We offer the following comments

- Using a price curve based on marginal costs does not restrict significantly the analysis and facilitates the economic dispatch in each case. Prices could be independent of marginal costs, then the analysis would complicate but conclusions would remain, although benefits would change. One must be careful in interpreting the action of "multiplication of the marginal cost"; it should be clear that in the paper only the slope is modified and factor b(i) is kept constant.
- When a perfect competitive market is in place, the developed model provides as a stable solution that where all agents ask for their marginal production costs (corresponding to the traditional economic dispatch solution). There is no need to go over this simulation to conclude that the short term marginal cost optimizes operation cost, maximizes income and net benefits [A]. On the short term, centralized dispatch at minimum cost is the economic optimal solution that makes public and private interests coincide. This is also valid for competitive wholesale hydrothermal electric energy markets, where dispatch minimizes expected operation and non served energy costs [B]. Nevertheless, this conclusion helps to validates the model.
- We tried to reproduce results reported in Table 2, and we were successful for cases where the three firms coordinate, but not for the case when A and B play against C. We obtained 20.47, -0.08 instead of 20.62, 0.30. Can the authors provide the marginal cost curves and check those numbers? The proposed benefit function is a relative one, it corresponds to the benefit achieved in comparison to the case where no exchanges are made with other sources, that is when local load is fed with own resources. If it is not beneficial, no deal would be achieved. Results in Table 2 indicate the benefits are positive for all players, which is in agreement with the above principle, but simulations should also identify cases where it is not convenient to negotiate and benefits are zero. In the indicated case where A and B offer their marginal costs and C offers a price that changes the slope of its marginal cost, company C would obtain a negative benefit, and therefore would not participate in the Pool. This is not reflected by numbers given in Table 2

- We suggest the following changes to the simulation of the games. When a player enters the game facing different alternatives and payments, it should first search secuencially for the dominated alternatives (those that in advance are *rationally not* eligible). If the equilibrium solution is not found by eliminating dominated strategies, then some selection criteria should be chosen to characterize the behavior of the player. Instead, the authors first define the selection criteria (minmax), before eliminating dominated strategies, which could lead to problems.

- The paper contributes conceptually to the identification of the perfect competitive market as well as possible problems with competition or entry barriers. It does not go into the negotiation process involved in distributing benefits within the coalition, which assumes will be formed if the benefits are greater than the sum of the individual earnings. Simulating such process would be very complex and is beyond the objectives of the paper. The discussion on how transmission capacity restrictions would provide incentives to break the coalition is interesting and should be explored further. Nevertheless, short term operations are not essentially modified, the difference being that in this case there will be different prices for different buses [C].

To apply the simulation to a real environment is a most difficult task. Conditions for perfect competition are highly dependent on the economic and technical condition of each electricity market and must be analyzed in each particular case. We clearly see differences in the Argentinean and Chilean markets, particularly given the location of hydroplants and the transmission restrictions. Besides, evaluating all possible strategies in a condition of imperfect competition would require an extraordinary computer task. In the Argentinean market, where 26 thermal generators and 13 hydroelectric ones compete, the maximum number of coalitions would be 2^{39}-1. Heuristic techniques could be developed based on operational experience to significantly reduce the number of coalitions to study. The comments of the authors on how to tackle this problem will be appreciated.

[A] Nishimura, F., Tabors, R.D., Ilic, M.D., Lacalle-Melero, J.R., "Benefit optimization of centralized and decentralized power systems in a multi-utility environment", IEEE Trans. Power Systems, Vol. 8, N 3, Aug. 1993, pp. 1180-1186

[B] Rudnick, H., Varela, H., Hogan, W., "Evaluation of alternatives for power system coordination and pooling in a competitive environment", IEEE/PES 1996 Winter Meeting, Baltimore, Md., January 21-25 1996

[C] Hogan, W. W., "Coordination for Competition in an Electricity Market", John F. Kennedy School of Government, Harvard University, Cambridge, Mass., March 2, 1995

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R.W. Ferrero, S.M. Shahidehpour and V.C. Ramesh--
The authors would like to thank the discussers for their interest in the paper. They have made valuable comments and raised important questions.

Transaction analysis in deregulated power systems is becoming an important issue in the electric industry. In this environment, the role of the regulator is not clearly defined yet. It appears to be clear that Pool regulations should promote competition among participants, while maximum system-wide benefits would be reached trough increased competition. We attempted to provide an analytical tool to systematically check the competitive conditions in deregulated power systems. In what follows, we try to answer the discussers' questions:

Dr. Jacobs

We see collusion as a consequence market conditions and regulations rather than a necessary condition for marginal cost pricing. Instead of analyzing Table 2 in the paper, we preferred to include the whole picture and analyze Table 3. We concluded that there is no incentive for participants to collude (that is, price it different than the marginal cost) as they get more benefits by pricing at marginal costs. The grand coalition is appealing to participants due to the threat of *potential* coalitions in the Pool. We agree with the discusser's final comment.

Drs. Rudnick, Watts and Moitre

The marginal cost curves can be obtained from Table 1 in the paper. We assumed that participants could supply their own load with local resources and interchange the excess power with the Pool. We checked the numbers in the paper and did not find any errors. The case indicated by the discussers corresponds to row "MM" in Table 2. Both A and B offer power at marginal costs and C varies its price between λ_0+ 1.7 c and λ_0+ 2.3 c.

However, the case pointed out by the discussers may happen in a different situation of participant's characteristics (cost, strategies, maximum and minimum power constraints, etc.). The proposed methodology is general enough to consider the proposed case. When a participant's benefit is negative for a combination of strategies, this situation will clearly have no chance to happen in the final transaction schedule. In this case, there is a signal to the participant to either change its price strategy accordingly or to decline participation in the Pool. The final characteristic functions for all possible coalitions will show which coalitions are likely to happen.

From numerical experiments performed during our study, we found that the max-min criteria lead to the same *pure-strategy* equilibrium as the iterative dominance procedure suggested by the reviewers. Using the max-min criteria we were able to find the possible *mixed-strategy equilibrium* in the game as well.

The computational requirements of the proposed procedure are high, however, as the discussers suggested, one can imagine heuristic techniques that will help reduce the computations. With respect to computational requirements we may add that,

- The game theoretic analysis is not to be applied to the "on line" dispatch. We may expect to perform the analysis for different situations in which calculation speed is not a critical issue.
- We can apply parallel processing to the proposed methodology. It is very easy to design an algorithm that takes advantage of the problem's structure. Transaction analyses for different combinations of strategies are independent events and can be solved simultaneously.

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